1)
```
π employee_id, name, expertise (
  assistant
  ⋈ assistant.boss = lecture.lectured_by (
    σ credits = 2 ∨ credits = 4 (lecture)
  )
)
```

2)
```
π student_id (
  σ title = 'Ethik' (lecture ⋈ attends)
)
∩
π student_id (
  σ title = 'Bioethik' (lecture ⋈ attends)
)
```

3)
```
π assistant.name (
  assistant
  ⋈ boss = professor.employee_id (
    σ professor.name = 'Kopernikus' (professor)
  )
)
```

4)
```
π lecture_id, title, credits (lecture)
-
π lecture_id, title, credits (
  lecture ⋈ lecture_id = successor (requires)
)
```

5)
```
π name (
  professor
  ⋈ employee_id = l_id (
    π L1.lectured_by → l_id (
      σ L1.lectured_by = L2.lectured_by ∧
        L2.lectured_by = L3.lectured_by ∧
        L1.lecture_id < L2.lecture_id ∧
        L2.lecture_id < L3.lecture_id (
          ρ L1 (lecture) × ρ L2 (lecture) × ρ L3 (lecture)
      )
    )
  )
)
```

6)
```
π student_id, name, semester (student)  -
π A1.student_id, A1.name, A1.semester (
  ρ A1 (student)
  ⋈ A1.semester < A2.semester
  ρ A2 (student)
)
```

7)
```
π student.name (
  student
  ⋈ student_id = sid (
    π student_id → sid, lecture_id (attends)
    ÷
    π lecture_id (σ lecture_id > 5050 (lecture))
  )
)
```

8)
```
π student_id (
  attends
  ⋈ lecture_id = successor (
    π successor (
      requires
      ⋈ predecessor = lecture_id (
        σ title = 'Grundzuege' (lecture)
      )
    )
  )
)
```